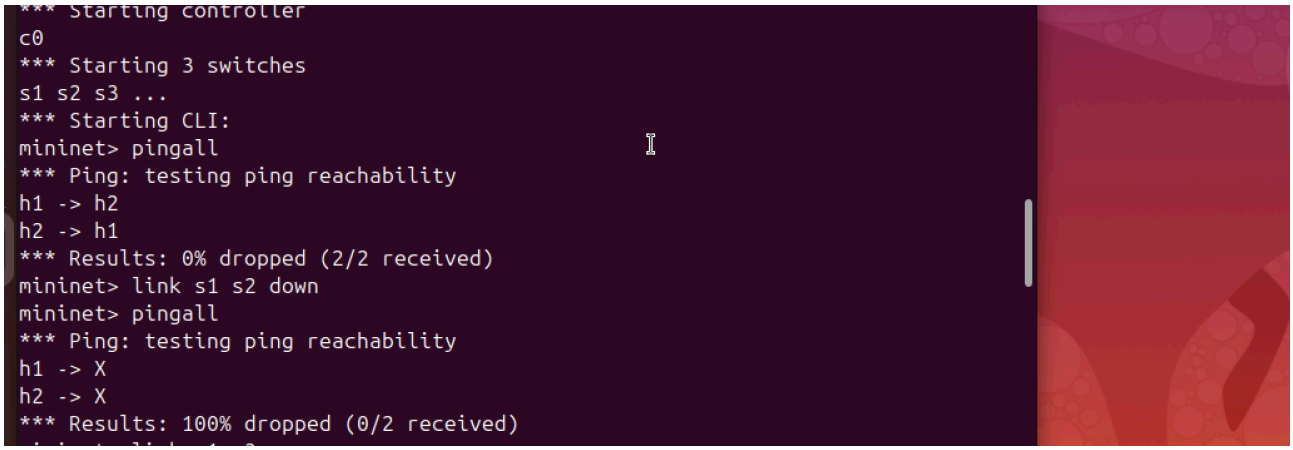
Link Failure Detection and Recovery using SDN

Detect link failures and dynamically update routing to restore connectivity using Software Defined Networking (SDN).

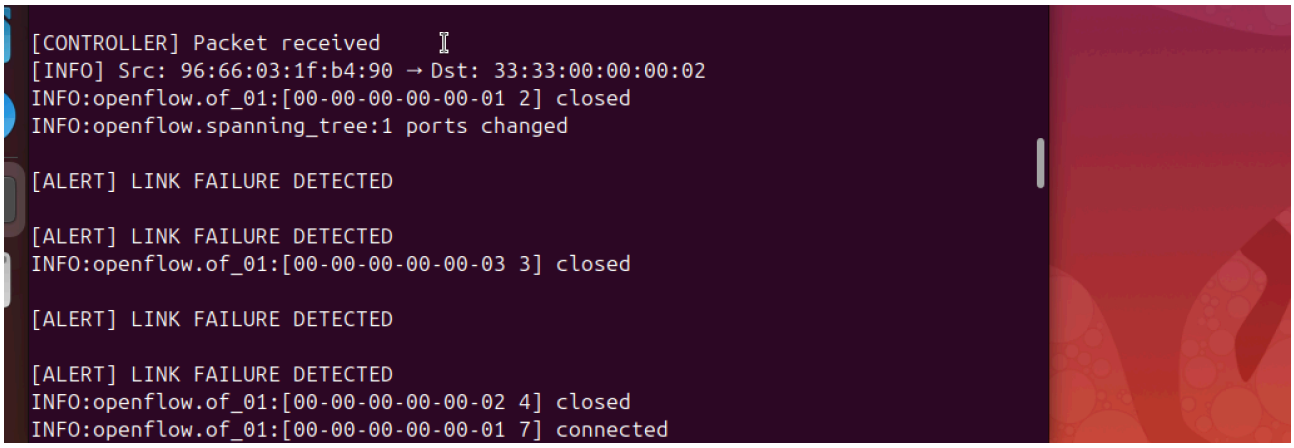
Project Overview

This project demonstrates how SDN enables adaptive networking by separating the control plane from the data plane. A centralized controller (POX) monitors the network topology and dynamically updates routing when a link failure occurs.

Screenshots
1.(Topology in readme file)
2.



3.



4.

```
mininet> link s1 s2 up
mininet> link s1 s3 up
mininet> link s3 s2 up
mininet> pingall
*** Ping: testing ping reachability
h1 -> *** Error: could not parse ping output: bash: hping: command not found

X
h2 -> h1
*** Results: 50% dropped (1/2 received)
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2
h2 -> h1
*** Results: 0% dropped (2/2 received)
mininet>
```